

1. Proof Insurance (Mandatory Class Proofs)

Theorem 1: Alternate Variance Formula

Proof. By definition of variance and utilizing the linearity properties of mathematical expectation ($\mathbb{E}$):

$$\begin{aligned}\text{Var}[X] &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X \cdot \mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[2X \cdot \mathbb{E}[X]] + \mathbb{E}[(\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X] \cdot \mathbb{E}[X] + (\mathbb{E}[X])^2 \quad (\text{Pull constants}) \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2.\end{aligned}\quad \square$$

Theorem 2: Generalized Conditioning Rule (Chain Rule)

Proof. Expand the Right-Hand Side (RHS) using conditional probability definition $P(A|B) = \frac{P(A \cap B)}{P(B)}$:

$$\begin{aligned}\text{RHS} &= P(E_1) \cdot \frac{P(E_1 \cap E_2)}{P(E_1)} \cdot \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_1 \cap E_2)} \cdots \\ &\quad \cdots \frac{P(E_1 \cap E_2 \cap \cdots \cap E_n)}{P(E_1 \cap E_2 \cap \cdots \cap E_{n-1})}\end{aligned}$$

Observe the telescoping cancellation where intermediate joint probability denominators cancel matching prior numerators:

$$\begin{aligned}\text{RHS} &= \cancel{P(E_1)} \cdot \frac{\cancel{P(E_1 \cap E_2)}}{\cancel{P(E_1)}} \cdots \frac{P(E_1 \cap \cdots \cap E_n)}{\cancel{P(E_1 \cap \cdots \cap E_{n-1})}} \\ &= P(E_1 \cap E_2 \cap \cdots \cap E_n) = \text{LHS}.\end{aligned}\quad \square$$

Bonus Proof Asset Block

Identity: $\mathbb{E}[cX] = c\mathbb{E}[X]$

Proof. $\mathbb{E}[cX] = \sum (cx)f(x) = c \sum xf(x) = c\mathbb{E}[X].$ $\square$

Identity: $\text{Var}[aX + b] = a^2 \text{Var}[X]$

Proof. Let $\mu = \mathbb{E}[X]$. $\mathbb{E}[aX + b] = a\mu + b$.
 $\text{Var}[aX + b] = \mathbb{E}[((aX + b) - (a\mu + b))^2]$
 $= \mathbb{E}[(aX - a\mu)^2] = \mathbb{E}[a^2(X - \mu)^2] = a^2 \text{Var}[X].$ $\square$

Identity: $P(A^c) = 1 - P(A)$

Proof. $A \cup A^c = S$ and $A \cap A^c = \emptyset$. By Axiom 3, $P(A \cup A^c) = P(A) + P(A^c) \implies P(S) = P(A) + P(A^c)$. By Axiom 2, $1 = P(A) + P(A^c) \implies P(A^c) = 1 - P(A)$. $\square$

Baseline Foundations & Set Axioms

The 3 Core Axioms of Probability:

- **Axiom 1:** $P(A) \geq 0$ (Probability cannot be negative)
- **Axiom 2:** $P(S) = 1$ (The entire sample space is certain)
- **Axiom 3:** If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$

Vital Derived Rules:

- **Complement Property:** $P(A^c) = 1 - P(A)$
- **Inclusion-Exclusion Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **DeMorgan's Laws:** $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$
- **Null Probability:** $P(\emptyset) = 0$

Boundary & Inequality Translation Vocabulary

Let X be a discrete process approximated by a Normal curve with parameters μ and σ .

- **"Fewer than k " / "Strictly less than k " ($X < k$):**
 Discrete Setup: $P(X \leq k - 1)$
 CLT Continuity Shift: $P(X_{\text{normal}} \leq k - 0.5)$
- **"At most k " / "No more than k " ($X \leq k$):**
 Discrete Setup: $P(X \leq k)$
 CLT Continuity Shift: $P(X_{\text{normal}} \leq k + 0.5)$
- **"More than k " / "Strictly greater than k " ($X > k$):**
 Discrete Setup: $1 - P(X \leq k)$
 CLT Continuity Shift: $P(X_{\text{normal}} \geq k + 0.5)$
- **"At least k " / "No fewer than k " ($X \geq k$):**
 Discrete Setup: $1 - P(X \leq k - 1)$
 CLT Continuity Shift: $P(X_{\text{normal}} \geq k - 0.5)$
- **"Between a and b inclusive" ($a \leq X \leq b$):**
 Discrete Setup: $P(X \leq b) - P(X \leq a - 1)$
 CLT Continuity Shift: $P(a - 0.5 \leq X_{\text{normal}} \leq b + 0.5)$

The Quick Rule of Thumb: If the integer value k is *included* in your target set, your continuous boundary must swell outward by 0.5 to capture that entire bar. If k is *excluded*, your boundary must shrink away from it by 0.5.

2. Integral Calculus Moment Toolbox (Section 5.1)

When processing continuous random variables, polynomials scaled by exponential decay profiles require precise integration setups:

Integration by Parts (IBP): $\int u dv = uv - \int v du$

Selection rule: Set Algebraic (x, x^2) to u , Exponential $(e^{-\lambda x})$ to dv .

- **Type A ($\mathbb{E}[X]$ Mean Framework):** $\int x e^{-\lambda x} dx$
Let $u = x \implies du = dx; dv = e^{-\lambda x} dx \implies v = -\frac{1}{\lambda} e^{-\lambda x}$.

$$\int x e^{-\lambda x} dx = -\frac{x}{\lambda} e^{-\lambda x} - \frac{1}{\lambda^2} e^{-\lambda x}$$

- **Type B ($\mathbb{E}[X^2]$ Second Moment Framework):** $\int x^2 e^{-\lambda x} dx$
Let $u = x^2 \implies du = 2x dx; dv = e^{-\lambda x} dx \implies v = -\frac{1}{\lambda} e^{-\lambda x}$.

$$\int x^2 e^{-\lambda x} dx = -\frac{x^2}{\lambda} e^{-\lambda x} + \frac{2}{\lambda} \int x e^{-\lambda x} dx$$

3. Master Scenario A: Polya Urn Sequence Transitions

Problem Pattern: Urn contains 2 Red, 3 Blue. Draw 3 times sequentially. Upon each selection, replace the drawn token *plus add 4 additional tokens of the opposite color*. Evaluate $P(R_1 | B_3)$.

Execution Workflow: Apply conditional definition: $P(R_1 | B_3) = \frac{P(R_1 \cap B_3)}{P(B_3)}$.

1. Track changing urn composition along paths starting with R_1 :

- Path $R_1 R_2 B_3$: $\frac{2}{5} \rightarrow$ Add 4 Blue (Urn: 2R, 7B) $\rightarrow \frac{2}{9} \rightarrow$ Add 4 Blue (Urn: 2R, 11B) $\rightarrow \frac{11}{13}$. Joint Path Value = $\frac{44}{585}$.
- Path $R_1 B_2 B_3$: $\frac{2}{5} \rightarrow$ (Urn: 2R, 7B) $\rightarrow \frac{7}{9} \rightarrow$ Add 4 Red (Urn: 6R, 7B) $\rightarrow \frac{7}{13}$. Joint Path Value = $\frac{98}{585}$.

2. Track changing urn composition along paths starting with B_1 :

- Path $B_1 R_2 B_3$: $\frac{3}{5} \rightarrow$ Add 4 Red (Urn: 6R, 3B) $\rightarrow \frac{6}{9} \rightarrow$ Add 4 Blue (Urn: 6R, 7B) $\rightarrow \frac{7}{13}$. Joint Path Value = $\frac{126}{585}$.
- Path $B_1 B_2 B_3$: $\frac{3}{5} \rightarrow$ (Urn: 6R, 3B) $\rightarrow \frac{3}{9} \rightarrow$ Add 4 Red (Urn: 10R, 3B) $\rightarrow \frac{3}{13}$. Joint Path Value = $\frac{27}{585}$.

3. Total Law of Probability (Denominator $P(B_3)$):

$$\frac{44+98+126+27}{585} = \frac{295}{585}.$$

4. Target Intersection Numerator ($P(R_1 \cap B_3)$): $\frac{44+98}{585} = \frac{142}{585}$.

5. Complete Division: $P(R_1 | B_3) = \frac{142/585}{295/585} = \frac{142}{295} \approx \mathbf{0.4814}$.

4. Continuous Variable Integrations (HW 10)

A variable is continuous if its CDF is continuous everywhere. Total area under the density curve must normalize to 1: $\int_{-\infty}^{\infty} f(x)dx = 1$.

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x)dx, \quad \mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 \cdot f(x)dx$$

5. Master Scenario B: Piecewise Running-Total CDFs

Problem Pattern: Solve for the normalization constant c and construct the absolute piecewise CDF expression $F(y)$ across all boundaries for:

$$f(y) = \begin{cases} cy^2 & \text{if } 0 \leq y < 2 \\ c(6-y) & \text{if } 2 \leq y \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

Execution Workflow:

1. Enforce standard area integration properties to isolate c :

$$\int_0^2 cy^2 dy + \int_2^6 c(6-y) dy = 1 \implies c \left[\frac{y^3}{3} \right]_0^2 + c \left[-\frac{(6-y)^2}{2} \right]_2^6 = 1$$

$$c \left(\frac{8}{3} - 0 \right) + c \left(0 - \left(-\frac{16}{2} \right) \right) = 1 \implies c \left(\frac{32}{3} \right) = 1 \implies c = \frac{3}{32}$$

2. Construct $F(y) = P(Y \leq y)$ by carrying forward accumulated area:

- For $y < 0$: $F(y) = 0$.
- For $0 \leq y < 2$: $F(y) = \int_0^y \frac{3}{32}t^2 dt = \frac{1}{32}y^3$.
- For $2 \leq y \leq 6$: Add the total historical area of the first piece ($F(2) = \frac{1}{32}(8) = \frac{1}{4}$) directly to the current running integral:

$$\begin{aligned} F(y) &= \frac{1}{4} + \int_2^y \frac{3}{32}(6-t) dt = \frac{1}{4} + \frac{3}{32} \left[6t - \frac{t^2}{2} \right]_2^y \\ &= \frac{1}{4} + \frac{3}{32} \left(\left(6y - \frac{y^2}{2} \right) - (12 - 2) \right) \\ &= -\frac{3}{64}y^2 + \frac{9}{16}y - \frac{11}{16} \end{aligned}$$

- For $y > 6$: $F(y) = 1$.

6. Master Scenario C: Multi-Tier Special Distributions

Problem Pattern: Individual frames fail independently with $p = 0.05$. A cycle flags the stream if it catches *at least 2 errors across a frame block of 40 elements*. Find the probability that the 4th block is the first one flagged.

Execution Workflow:

1. Solve inner probability p^* using Binomial distribution ($n = 40, p = 0.05$):

$$p^* = P(X \geq 2) = 1 - P(X \leq 1) = 1 - \text{binomcdf}(40, 0.05, 1) \approx 0.6010$$

2. Solve outer probability using Geometric distribution ($k = 4, p^* \approx 0.6010$):

$$P(Y = 4) = (1 - p^*)^3 \cdot p^* = (0.3990)^3 \cdot 0.6010 = \mathbf{0.0382}$$

7. Master Scenario D: CLT Continuity Adjustments

Problem Pattern: An application has an error rate of $p = 0.20$. Across 900 independent packet items, evaluate the probability that the final error count falls *between 170 and 195 inclusive*.

Execution Workflow:

1. Local parameters: $\mu = np = 900(0.20) = 180$.
 $\sigma^2 = np(1-p) = 900(0.20)(0.80) = 144 \implies \sigma = 12$.
2. Shift discrete integer boundaries outwards by ± 0.5 to account for histogram bar width under continuous normal models:

$$P(170 \leq X \leq 195) \approx P(169.5 \leq X_{\text{normal}} \leq 195.5)$$

3. Convert to standard Z-scores:

$$Z_{\text{low}} = \frac{169.5 - 180}{12} = -0.88; \quad Z_{\text{high}} = \frac{195.5 - 180}{12} = 1.29$$

4. Final area score: $\text{normalcdf}(-0.88, 1.29, 0, 1) = \mathbf{0.7121}$.

8. Core Linear & Operational Shortcuts

- **Set Axioms (HW 1–2):** DeMorgan's: $(A \cup B)^c = A^c \cap B^c$.
Inclusion-Exclusion: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **Linear Operations (Section 4.2–4.3):** Expectation is linear: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$. Variance scales by the square and drops translation constants entirely: $\text{Var}[aX + b] = a^2\text{Var}[X]$.
- **Staircase CDF (Section 4.1):** Discrete step height jump sizes indicate localized point probability values: $P(X = x_i) = F(x_i) - F(x_{i-})$.